

Unit 3.6 Logarithmic differentiation .

Note Title

2012/04/16

In this unit:

$$\bullet \frac{d}{dx} \ln x$$

$$\bullet \frac{d}{dx} \ln |x|$$

$$\bullet \frac{d}{dx} x^{\sin x}$$

$$\bullet \frac{d}{dx} \frac{\sin x \cdot e^x \cdot \cos 2x}{x^3 \sqrt{1+x^2}}$$

First:

$$y = \ln x$$

$$\Rightarrow e^y = x$$

$$\Rightarrow e^y \cdot \frac{dy}{dx} = 1$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{e^y} = \frac{1}{x}.$$

$$\begin{array}{c} f(x) \\ e^{\uparrow} = x \\ e^{f(x)} \cdot f'(x) \end{array}$$

$$\text{Eg. } \frac{d}{dx} e^{2x} \cdot \ln x = e^{2x} \cdot 2 \cdot \ln x + e^{2x} \cdot \frac{1}{x}.$$

$$\text{Also: } \frac{d}{dx} \ln(f(x)) = \frac{f'(x)}{f(x)} \quad \text{NB.}$$

Examples Find $\frac{dy}{dx}$ if

$$1. \frac{d}{dx} \ln(x^3 + 2) = \frac{3x^2}{x^3 + 2}$$

$$2. \frac{d}{dx} \ln(\sin x) = \frac{\cos x}{\sin x}$$

$$3. \frac{d}{dx} \ln(x^2 + 1) \cdot \tan x = \frac{2x}{x^2 + 1} \cdot \tan x + \ln(x^2 + 1) \cdot \sec^2 x$$

$$4. \quad \underline{x \cdot \ln y = 4.}$$
$$\ln y + x \cdot \left(\frac{1}{y} \cdot \frac{dy}{dx} \right) = 0$$

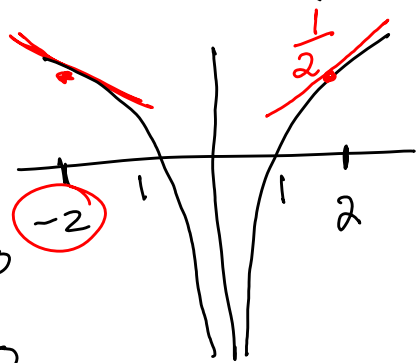
$$\Rightarrow \frac{dy}{dx} = -\ln y \times \frac{y}{x} = \frac{-y \ln y}{x}$$

Next: $\frac{d}{dx} \ln|x|$

$$\ln|x| = \begin{cases} \ln x & \text{if } x > 0 \\ \ln(-x) & \text{if } x < 0 \end{cases}$$

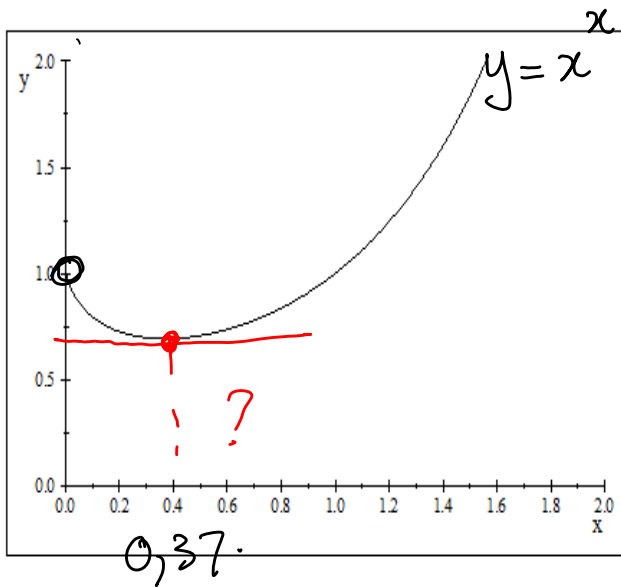
$$\frac{d}{dx} \ln|x| = \begin{cases} \frac{1}{x} & \text{if } x > 0 \\ \frac{-1}{-x} & \text{if } x < 0. \end{cases}$$

$$\Rightarrow \frac{d}{dx} \ln|x| = \frac{1}{x}, \quad x \neq 0,$$



Derivative of $y = f(x)g(x)$

Eg. Find $\frac{dy}{dx}$ if $y = x^x$



TAKE NOTE.

number
 $y = x^{10}$
 $y' = 10x^9$

~~$y = x^x$
 $y' = x \cdot x^{x-1}$~~

$$\begin{aligned}
 y &= x^x \\
 \Rightarrow \ln y &= \ln x^x \\
 \Rightarrow \ln y &= x \ln x \\
 \Rightarrow \frac{d}{dx} \ln y &= \frac{d}{dx} x \cdot \ln x \\
 \Rightarrow \frac{1}{y} \cdot \frac{dy}{dx} &= \ln x + 1 \\
 \Rightarrow \frac{dy}{dx} &= y (\ln x + 1) \\
 &= x^x (\ln x + 1)
 \end{aligned}$$

Where is $\frac{dy}{dx} = 0$?

$$\Rightarrow x^x (\underbrace{\ln x + 1}_0) = 0$$

$$\Rightarrow \ln x = -1$$

$$\Rightarrow x = e^{-1} = \frac{1}{e} = 0,37$$

$$\ln 2 = 0,69$$

$$\frac{1}{6} = 0,17$$

$$\sqrt{2} = 1,414$$

Eg. Find $\frac{dy}{dx}$ if $y = x^{\sin x}$

$$\ln y = \ln x^{\sin x}$$

$$\Rightarrow \ln y = \sin x \cdot \ln x$$

$$\Rightarrow \frac{1}{y} \cdot \frac{dy}{dx} = \cos x \ln x + \frac{\sin x}{x}$$

$$\Rightarrow \frac{dy}{dx} = x^{\sin x} \left(\cos x \ln x + \frac{\sin x}{x} \right)$$

Finally a "party trick"

Find $\frac{dy}{dx}$ if $y = \frac{\sin x \cdot e^x}{x^3 \sqrt{1+x^2}}$

$$\Rightarrow \ln y = \ln \sin x + \ln e^x - \ln x^3 - \ln \sqrt{1+x^2}$$
$$= \ln \sin x + x - 3 \ln x - \frac{1}{2} \ln(1+x^2)$$

$$\Rightarrow \frac{1}{y} \cdot \frac{dy}{dx} = \frac{\cos x}{\sin x} + 1 - \frac{3}{x} - \frac{1}{2} \frac{2x}{1+x^2}$$

$$\Rightarrow \frac{dy}{dx} = \underbrace{\frac{\sin x \cdot e^x}{x^3 \sqrt{1+x^2}}}_y \left(\frac{\cos x}{\sin x} + 1 - \frac{3}{x} - \frac{x}{1+x^2} \right)$$

Another eg:

Find $\frac{dy}{dx}$ if $y = \frac{(\tan x)^3}{\sqrt{\sin x} \cdot x^{10}}$

$$\Rightarrow \frac{dy}{dx} = \frac{(\tan x)^3}{\sqrt{\sin x} \cdot x^{10}} \left(\frac{3 \sec^2 x}{\tan x} - \frac{1}{2} \frac{\cos x}{\sin x} - \frac{10}{x} \right)$$